

PROJECT REPORT

CUSTOMER CHURN ANALYSIS DASHBOARD & INSIGHTS REPORT

Internship
Data Science and Analytics
Organization
Future Interns

Prepared by
Amirtha Rithika P

SUMMARY

This project focuses on analyzing customer churn using a telecom subscription dataset to understand why customers leave and what factors influence retention. The analysis aims to identify churn patterns across different customer segments, service types, and payment behaviors. Using Python (Google Colab) and Microsoft Excel, data was cleaned, analyzed, and visualized through an interactive dashboard. The project provides meaningful insights that help businesses reduce churn and improve customer retention strategies.

PROBLEM STATEMENT

Organizations often struggle to retain customers due to a lack of clear insights into customer behavior. This leads to challenges such as:

- Inability to identify customers at risk of leaving
- Lack of understanding of churn drivers
- Poor customer retention strategies

This project addresses these challenges by analyzing customer data to uncover churn patterns and presenting insights through visualizations and dashboards.

PROJECT OBJECTIVES

- Analyze customer churn patterns
- Identify high-risk customer segments
- Evaluate impact of contract types on churn
- Analyze tenure and customer lifetime behavior
- Build an interactive churn dashboard
- Provide actionable business recommendations

DATA DESCRIPTION

The dataset used in this project is the Telco Customer Churn dataset, which contains detailed information about customer subscriptions and behavior.

It consists of approximately 7,000+ records with multiple features representing customer demographics, services, and account information.

Key attributes include:

- CustomerID – Unique identifier
- Tenure – Number of months customer stayed
- MonthlyCharges – Monthly subscription cost
- TotalCharges – Total amount paid
- Contract – Type of subscription (Month-to-month, One year, Two year)
- InternetService – Type of internet service
- PaymentMethod – Mode of payment
- Churn – Indicates whether customer left (Yes/No)

The dataset includes both numerical and categorical variables. Monthly charges range from low to high values, while tenure varies from new to long-term customers. The churn variable provides a clear indication of customer retention behavior.

DATA PREPROCESSING

Tools & Technologies Used:

- Google Colab (Python – Pandas, Matplotlib, Seaborn)
- Microsoft Excel (Pivot Tables, Charts, Slicers)

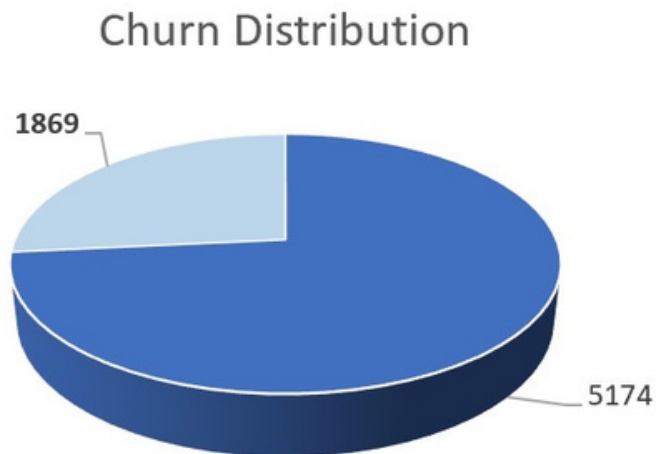
Steps Performed:

- Loaded dataset and explored structure
- Converted TotalCharges to numeric format
- Handled missing values using median imputation
- Converted Churn into numerical format (0 and 1)
- Created new feature:
 - Tenure Group (0–1 Year, 1–2 Years, 2+ Years)
- Structured dataset for Excel analysis
- Prepared data for Pivot Table and dashboard creation

Data preprocessing ensured the dataset was clean, consistent, and suitable for analysis. Missing values and incorrect data types were handled, and new features were created to improve analysis quality.

EXPLORATORY DATA ANALYSIS (EDA)

1. Churn Distribution



This chart shows the proportion of customers who stayed versus those who churned. It provides an overall understanding of the churn rate in the business.

Insights:

- A significant portion of customers have churned
- Retained customers still form the majority
- Indicates the need for retention strategies

EXPLORATORY DATA ANALYSIS (EDA)

2. Churn by Contract



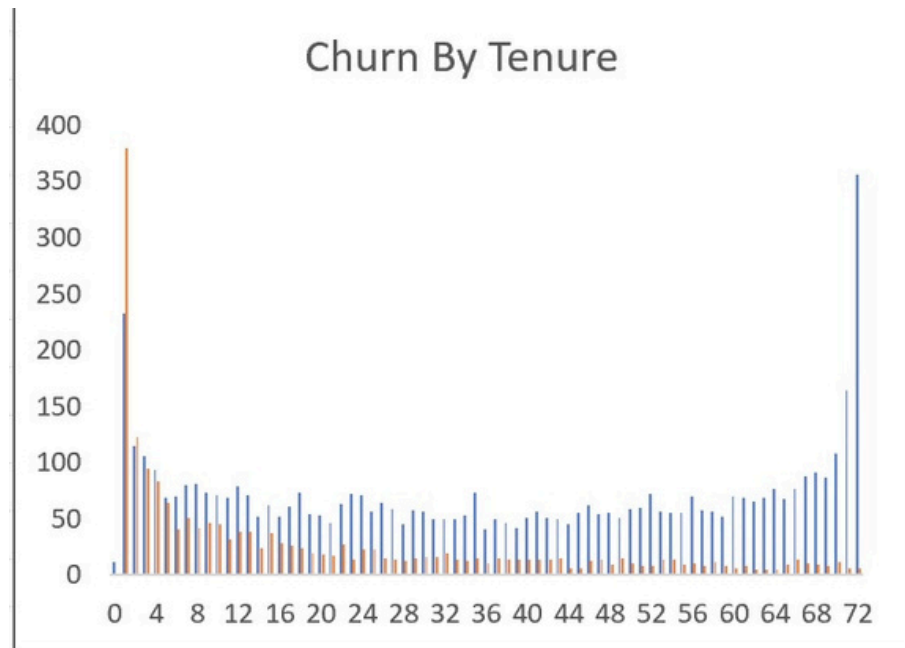
This chart compares churn across different contract types. It helps identify how customer commitment impacts churn behavior.

Insights:

- Month-to-month contracts show the highest churn
- Long-term contracts (1 year, 2 year) have lower churn
- Contract type strongly influences customer retention

EXPLORATORY DATA ANALYSIS (EDA)

3. Churn by Tenure



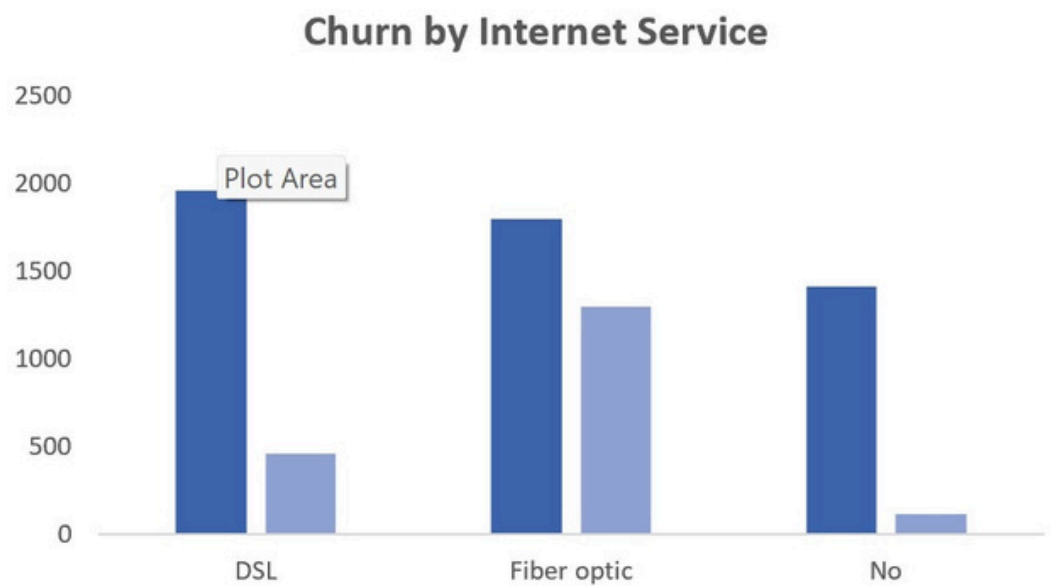
This chart analyzes churn based on how long customers have stayed with the company.

Insights:

- Customers with tenure less than 1 year have the highest churn
- Churn decreases as tenure increases
- Long-term customers are more stable

EXPLORATORY DATA ANALYSIS (EDA)

4.Churn by Internet Service



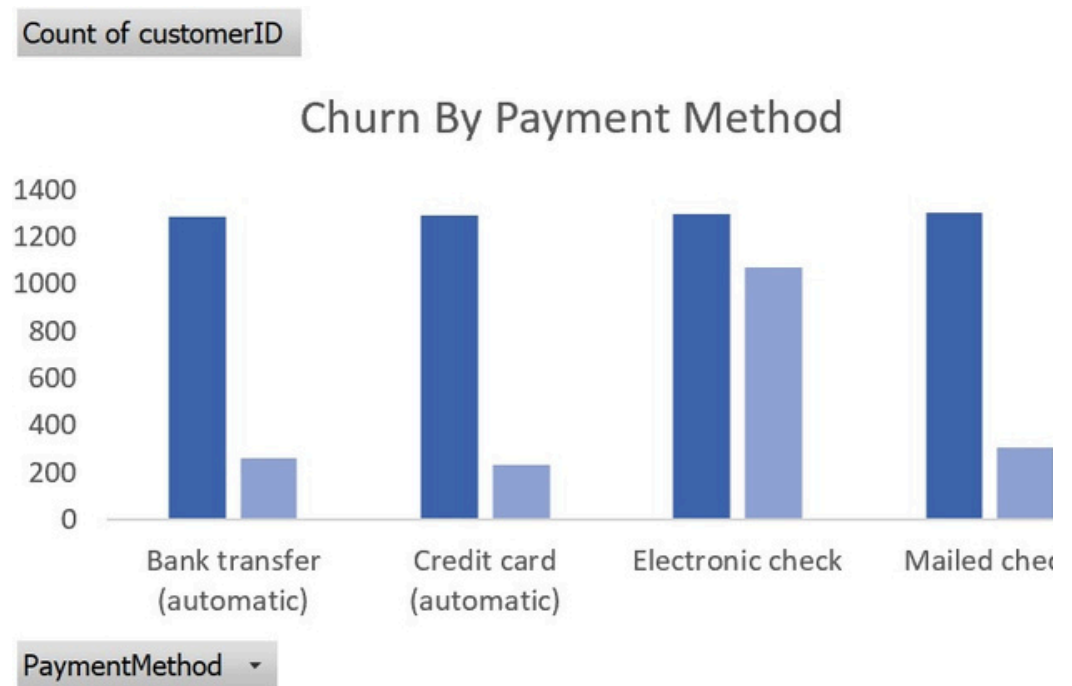
This chart shows churn distribution across different internet service types.

Insights:

- Fiber optic users have higher churn rates
- DSL users show moderate churn
- Customers with no internet service have lowest churn

EXPLORATORY DATA ANALYSIS (EDA)

5. Churn by Payment Method

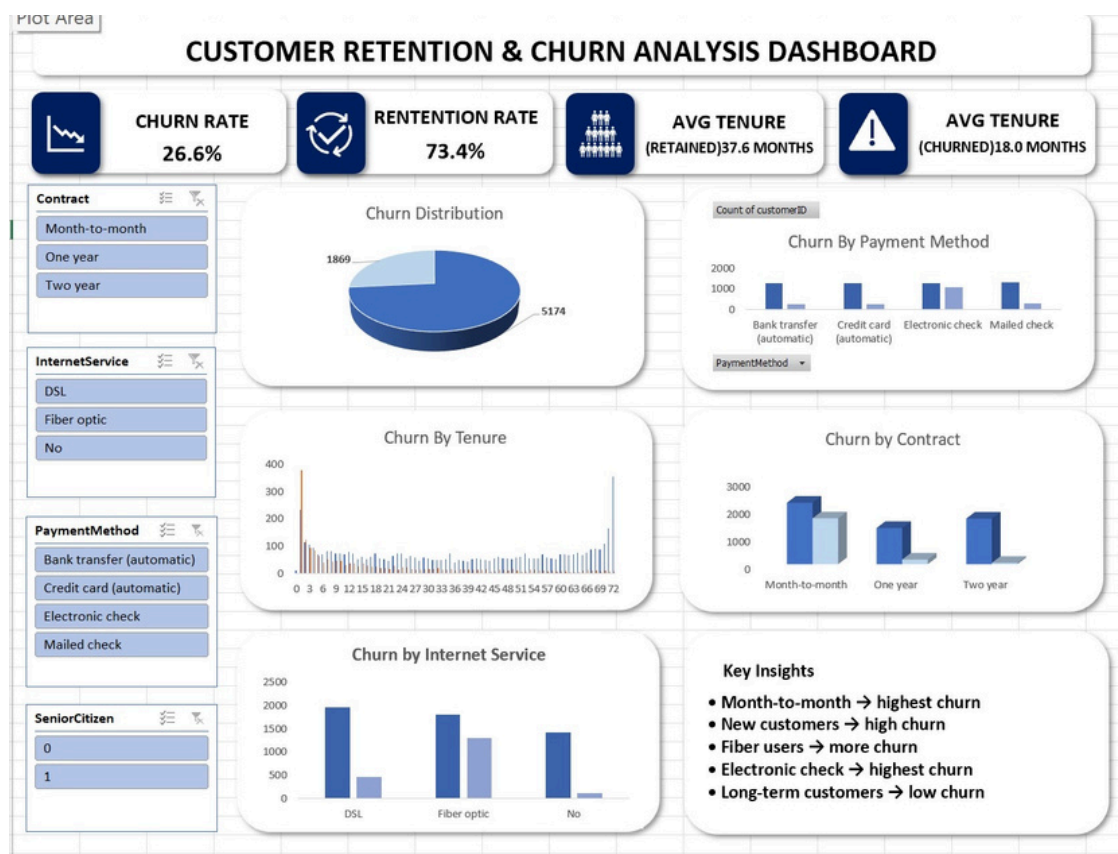


This chart compares churn based on different payment methods used by customers.

Insights:

- Electronic check users show the highest churn
- Automatic payment methods (credit card, bank transfer) have lower churn
- Payment behavior influences customer retention

DASHBOARD



The dashboard provides a clear and interactive overview of customer churn analysis using key visualizations. It includes charts for Churn Distribution, Churn by Contract, Churn by Tenure, Churn by Internet Service, and Churn by Payment Method to highlight different aspects of customer behavior. Users can easily analyze which customer segments are most likely to churn and how different factors influence retention.

Slicers are used to filter data by Contract, Internet Service, Payment Method, and Gender, allowing dynamic and real-time interaction with the dashboard. The layout is clean and user-friendly, making it easy to compare patterns and identify trends. Overall, the dashboard acts as a powerful decision-making tool, helping businesses understand churn behavior and improve retention strategies.

KEY INSIGHTS

- Month-to-month contracts show the highest churn, indicating low commitment
- New customers (0–1 year) churn the most, showing early dissatisfaction
- Fiber optic users have higher churn, possibly due to pricing or service issues
- Electronic check users show the highest churn, indicating low loyalty
- Long-term and auto-payment customers are more stable and less likely to churn

BUSINESS RECOMMENDATIONS

- Encourage customers to switch to long-term contracts
- Improve onboarding experience for new customers
- Review pricing and service quality for fiber plans
- Promote automatic payment methods
- Implement retention strategies for high-risk customers

CONCLUSION

This project demonstrates how customer churn analysis can help businesses understand customer behavior and improve retention strategies. By analyzing key factors such as contract type, tenure, service usage, and payment methods, meaningful insights were generated. The interactive dashboard simplifies complex data into an easy-to-understand format, enabling businesses to make informed and data-driven decisions to reduce churn and improve customer loyalty.

THANKYOU